Soundness of Type Difference in SKJS

Abstract

We prove that the type difference (diff) operation in SKJS satisfies properties ensuring sound type inference.

1 Definitions Index

Symbol	Defined In
Tags	Definition 1
Payload, Val, rtag	Definition 2
SType	Definition 3
tag	Definition 4
A0	Definition 5
A1	Definition 6
STypes, tags$(\cdot)$	Definition 7
STC$(\cdot,\cdot)$	Definition 19
TInst	Definition 8
TInsts	Definition 9
TParam	Definition 10
Clause	Definition 11
Clauses	Definition 12
TType	Definition 13
diff (soundness P1/P2)	Definition 14

2 Type Representation

Definition 1 (Runtime Tags)**.** *Let* Tags *be the set of runtime tags (the coarse runtime kind of a value). In the current implementation, tags are represented as integers, so one may take* $\mathsf{Tags} = \{0, 1, \ldots, 11\}$.

Definition 2 (Runtime Values and Tags)**.** *Assume runtime values carry their tag as part of the representation: let* Payload *be an abstract set of payloads and define*

$$\mathsf{Val} = \mathsf{Tags} \times \mathsf{Payload}.$$

Define the runtime tag function $\mathsf{rtag} : \mathsf{Val} \to \mathsf{Tags}$ *as the first projection:*

$$\mathsf{rtag}((k, p)) = k.$$

Definition 3 (Simple Types)**.** *Let* SType *be the set of* simple types*, i.e. the atomic type forms that have a single runtime tag. Formally,* SType *is the smallest set generated by the following constructors:*

- *primitive atoms:* $\texttt{undefined}$, $\texttt{void}$, $\texttt{null}$, $\texttt{boolean}$, $\texttt{number}$, $\texttt{string}$, $\texttt{bigint}$
- *boolean literals:* $\texttt{true}$, $\texttt{false}$
- *numeric literals:* n *for* $n \in \mathbb{R}$
- *string literals: "x" for strings* x
- *structured atoms:* $\texttt{tuple}(\ldots)$, $\texttt{Promise}(\ldots)$, $\texttt{object}(\ldots)$, $\texttt{algebraic}(\ldots)$ *(tagged unions/ADTs with a runtime discriminant),* $\texttt{function}(\ldots)$

We write these constructors suggestively; their internal parameters are left abstract, as they are not needed for the tag-level reasoning in this note.

Definition 4 (Simple Types and Tags)**.** *The function* $\mathsf{tag} : \mathsf{SType} \to \mathsf{Tags}$ *assigns tags:*

$$\begin{aligned}
\mathsf{tag}(\texttt{undefined}) &= 0 & \mathsf{tag}(\texttt{void}) &= 1 \\
\mathsf{tag}(\texttt{null}) &= 2 & \mathsf{tag}(\texttt{boolean}), \mathsf{tag}(\texttt{true}), \mathsf{tag}(\texttt{false}) &= 3 \\
\mathsf{tag}(\texttt{number}), \mathsf{tag}(n) &= 4 \text{ for } n \in \mathbb{R} & \mathsf{tag}(\texttt{string}), \mathsf{tag}(\text{“}x\text{”}) &= 5 \\
\mathsf{tag}(\texttt{bigint}) &= 6 & \mathsf{tag}(\texttt{tuple}(\ldots)) &= 7 \\
\mathsf{tag}(\texttt{Promise}(\ldots)) &= 8 & \mathsf{tag}(\texttt{object}(\ldots)) &= 9 \\
\mathsf{tag}(\texttt{algebraic}(\ldots)) &= 10 & \mathsf{tag}(\texttt{function}(\ldots)) &= 11
\end{aligned}$$

The semantic interpretation $[\![\cdot]\!] : \mathsf{SType} \to \mathcal{P}(\mathsf{Val})$ *maps each simple type to its set of inhabitants.*

Note. *We do* not *define* $[\![\cdot]\!]$ *in this document; it is treated as an abstract interpretation function. Instead, we state explicit assumptions about its behavior as axioms (A0–A1).*

Definition 5 (**A0** (Runtime Tag Consistency))**.** *The semantic interpretation of simple types agrees with runtime tags:*

$$\forall s \in \mathsf{SType}.\ \forall v \in \mathsf{Val}.\ v \in [\![s]\!] \implies \mathsf{rtag}(v) = \mathsf{tag}(s).$$

Definition 6 (**A1** (Tag Disjointness))**.**

$$\forall s_1, s_2 \in \mathsf{SType}.\ \mathsf{tag}(s_1) \neq \mathsf{tag}(s_2) \implies [\![s_1]\!] \cap [\![s_2]\!] = \emptyset$$

Definition 7 (STypes)**.** STypes *is a finite partial map* $\mathsf{Tags} \rightharpoonup \mathsf{SType}$ *with at most one entry per tag: if* $S(k) = s$ *then* $\mathsf{tag}(s) = k$*.*

Partial-map conventions. *We write* $\mathrm{dom}(S)$ *for the domain of defined keys. When a definition returns "undefined" at a key* k*, that means precisely* $k \notin \mathrm{dom}(S)$ *(i.e. the partial map has no entry at* k*).*

$$[\![S]\!] = \bigcup_{k \in \mathrm{dom}(S)} [\![S(k)]\!]$$

We write $\mathsf{tags}(S) = \mathrm{dom}(S)$*. We write* $\{k \mapsto s\}$ *for the singleton partial map with domain* $\{k\}$*.*

Intuition. *An* STypes *value represents the* tag-level *part of a union type: it records, for each runtime tag k that may occur, a single representative simple type s of that tag. The map shape enforces a normalization invariant ("at most one entry per tag"), which matches the implementation: many operations reason by tags and intentionally do not keep multiple alternatives of the same tag.*

Example. *The union* `number | string` *is represented as:*

$$S = \{4 \mapsto \texttt{number},\ 5 \mapsto \texttt{string}\}$$

and has semantics $[\![S]\!] = [\![\texttt{number}]\!] \cup [\![\texttt{string}]\!]$. *In contrast,* `number` *and the numeric literal* `1` *share the same tag* 4, *so an* STypes *value cannot simultaneously store both; this is why, for example, a tag-based "diff" cannot precisely express* `number \ 1`.

Definition 8 (Type Instances)**.** *Let* TInst *be the set of type instances (type variables or generic instantiations).*

Examples. *A type variable instance is something like T (a unification variable introduced during type checking; this is* not *a source-level type parameter). A generic instantiation instance is a named type applied to type arguments, e.g.* `Promise`$\langle$`number`$\rangle$ *or* `Array`$\langle$`string`$\rangle$.

Each instance $i \in$ TInst has a semantic interpretation $[\![i]\!] \subseteq$ Val.

Definition 9 (Type Instance Sets)**.** *Let* $\mathsf{TInsts} = \mathcal{P}_{\text{fin}}(\mathsf{TInst})$ *be the set of* finite *sets of instances. For $I \in$ TInsts:*

$$[\![I]\!] = \bigcup_{i \in I} [\![i]\!].$$

Definition 10 (Type Parameters)**.** *Let* TParam *be the set of* type parameters *(generic binders), i.e. names introduced by source-level generic declarations such as* `function f`$\langle T \rangle(\ldots)$ *or* `type Box`$\langle T \rangle = \ldots$. *They differ from type-variable instances: type parameters are binders whose meaning is fixed by the caller/instantiation, whereas type-variable instances are inference unknowns whose meaning is solved by the type checker.*

Definition 11 (Clause)**.** *A clause* $c = (\mathsf{vars}, \mathsf{stypes})$ *consists of:*

- $\mathsf{vars} \subseteq$ TParam*: a finite set of type parameters*
- $\mathsf{stypes} \in \{\mathsf{None}\} \cup \{\mathsf{Some}(S) \mid S \in \mathsf{STypes}\}$

Each type parameter $v \in$ TParam has an interpretation $[\![v]\!] \subseteq$ Val. Semantics:

$$[\![(\mathsf{vars}, \mathsf{stypes})]\!] = \left(\bigcap_{v \in \mathsf{vars}} [\![v]\!] \right) \cap [\![\mathsf{stypes}]\!]$$

where $[\![\mathsf{None}]\!] = \mathsf{Val}$ *(top) and* $[\![\mathsf{Some}(S)]\!] = [\![S]\!]$. *For* $\mathsf{vars} = \emptyset$, $\bigcap_{v \in \emptyset} [\![v]\!] = \mathsf{Val}$.

Well-formedness: $\mathsf{stypes} = \mathsf{None} \implies \mathsf{vars} = \emptyset$ *(i.e., $(\emptyset, \mathsf{None})$ is the unique top clause).*

Definition 12 (Clauses)**.** Clauses *is a finite map from* $\mathcal{P}_{\text{fin}}(\mathsf{TParam})$ *to* Clause*, with at most one clause per key, and such that keys agree with clause variables: if $C(k) = (\mathsf{vars}, \mathsf{stypes})$ then* $\mathsf{vars} = k$.

$$[\![C]\!] = \bigcup_{k \in \mathrm{dom}(C)} [\![C(k)]\!]$$

Example. *Let* $S_{\mathsf{num}} = \{4 \mapsto \texttt{number}\}$ *and* $S_{\mathsf{str}} = \{5 \mapsto \texttt{string}\}$. *Let* $A, B \in \mathsf{TParam}$ *be two type parameters. Consider the clauses map* C *with two entries:*

$$C(\{A\}) = (\{A\}, \mathsf{Some}(S_{\mathsf{num}})) \qquad C(\{B\}) = (\{B\}, \mathsf{Some}(S_{\mathsf{str}})).$$

Then

$$[\![C]\!] = \big([\![A]\!] \cap [\![\texttt{number}]\!]\big) \cup \big([\![B]\!] \cap [\![\texttt{string}]\!]\big).$$

To illustrate a non-singleton key, consider instead the single-entry map C' *defined by:*

$$C'(\{A, B\}) = (\{A, B\}, \mathsf{Some}(S_{\mathsf{num}})).$$

Then $[\![C']\!] = [\![A]\!] \cap [\![B]\!] \cap [\![\texttt{number}]\!]$, *i.e. the clause requires* both *parameters.*

Definition 13 (TType)**.** $\mathsf{TType} = (\mathsf{insts}, \mathsf{clauses})$ *where* $\mathsf{insts} \in \mathsf{TInsts}$ *and* $\mathsf{clauses} \in \mathsf{Clauses}$.

$$[\![(\mathsf{insts}, \mathsf{clauses})]\!] = [\![\mathsf{insts}]\!] \cup [\![\mathsf{clauses}]\!]$$

Schematic. *It is helpful to keep the following picture in mind:*

$$\underbrace{(\mathsf{insts}, \mathsf{clauses})}_{\mathsf{TType}} \quad = \quad \underbrace{\mathsf{insts}}_{\text{“instance” part}} \quad \cup \quad \underbrace{\mathsf{clauses}}_{\text{“conditional” part}}$$

and later $\mathsf{diff}_{\mathsf{TType}}$ *simply applies* diff *componentwise to these two parts.*

Example. *Let* $I = \{\texttt{Promise}\langle\texttt{number}\rangle\} \in \mathsf{TInsts}$ *and let* C *be the clauses map from the previous example. Then* $t = (I, C)$ *denotes the type whose semantics is:*

$$[\![t]\!] = [\![\texttt{Promise}\langle\texttt{number}\rangle]\!] \cup [\![C]\!].$$

3 The Diff Operation

Definition 14 (Diff Soundness Properties)**.** *Let* X *be a set equipped with a semantic interpretation* $[\![\cdot]\!] : X \to \mathcal{P}(\mathsf{Val})$. *An operation* $\mathsf{diff} : X \times X \to X$ *is* sound *(with respect to* $[\![\cdot]\!]$*) if for all* $x_1, x_2 \in X$, *letting* $x_3 = \mathsf{diff}(x_1, x_2)$, *the following properties hold:*

(P1) $[\![x_3]\!] \subseteq [\![x_1]\!]$ (under-approximation)

(P2) $[\![x_3]\!] \cup [\![x_2]\!] \supseteq [\![x_1]\!]$ (coverage)

In this document we instantiate X *with* STypes, Clause, $\mathsf{Clauses}$, *and* TType.

Reader's guide. *These two properties imply that* $[\![x_3]\!]$ *is always a refinement of* $[\![x_1]\!]$, *and that it still covers all of* $[\![x_1]\!]$ *once* $[\![x_2]\!]$ *is added back. Equivalently (Lemma 15),* $[\![x_3]\!]$ *is an* over-approximation *of the true set difference* $[\![x_1]\!] \setminus [\![x_2]\!]$ *while staying within* $[\![x_1]\!]$.

Lemma 15 (Equivalent Characterization)**.** *For all* $x_1, x_2, x_3 \in X$, *(P1)* $\wedge$ *(P2) holds if and only if*

$$[\![x_1]\!] \setminus [\![x_2]\!] \subseteq [\![x_3]\!] \subseteq [\![x_1]\!].$$

Proof. ($\Rightarrow$) Let $v \in [\![x_1]\!] \setminus [\![x_2]\!]$. By (P2), $v \in [\![x_3]\!] \cup [\![x_2]\!]$. Since $v \notin [\![x_2]\!]$, we have $v \in [\![x_3]\!]$. Thus $[\![x_1]\!] \setminus [\![x_2]\!] \subseteq [\![x_3]\!]$. Combined with (P1): $[\![x_1]\!] \setminus [\![x_2]\!] \subseteq [\![x_3]\!] \subseteq [\![x_1]\!]$.

($\Leftarrow$) The right-hand inclusion $[\![x_3]\!] \subseteq [\![x_1]\!]$ is exactly (P1). For (P2),

$$[\![x_3]\!] \cup [\![x_2]\!] \supseteq ([\![x_1]\!] \setminus [\![x_2]\!]) \cup [\![x_2]\!] \supseteq [\![x_1]\!].$$

□

Definition 16 (Completeness)**.** *A diff operation is* complete *if:*

$$[\![\mathsf{diff}(t_1, t_2)]\!] = [\![t_1]\!] \setminus [\![t_2]\!]$$

By the equivalent characterization, (P1) ∧ (P2) implies $[\![t_1]\!] \setminus [\![t_2]\!] \subseteq [\![\mathsf{diff}(t_1, t_2)]\!]$*. Completeness additionally requires the reverse:* $[\![\mathsf{diff}(t_1, t_2)]\!] \subseteq [\![t_1]\!] \setminus [\![t_2]\!]$*.*

3.1 STypes Diff

Definition 17 (STypes Diff)**.**

$$\mathsf{diff}_{\mathsf{STypes}}(L, R)(k) = \begin{cases} L(k) & \text{if } k \in \mathsf{tags}(L) \setminus \mathsf{tags}(R) \\ \text{undefined} & \text{otherwise} \end{cases}$$

Intuition. *Since an* STypes *value stores at most one representative per runtime tag, this diff only removes whole tags: it preserves exactly the tags present on the left but absent on the right.*

Lemma 18 (STypes Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] \subseteq [\![L]\!]$*.*

Proof. By definition, $\mathsf{tags}(\mathsf{diff}_{\mathsf{STypes}}(L, R)) = \mathsf{tags}(L) \setminus \mathsf{tags}(R) \subseteq \mathsf{tags}(L)$, and for all $k \in \mathsf{tags}(\mathsf{diff}_{\mathsf{STypes}}(L, R))$ we have $\mathsf{diff}_{\mathsf{STypes}}(L, R)(k) = L(k)$. Therefore:

$$[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] = \bigcup_{k \in \mathsf{tags}(\mathsf{diff}_{\mathsf{STypes}}(L,R))} [\![L(k)]\!] \subseteq \bigcup_{k \in \mathsf{tags}(L)} [\![L(k)]\!] = [\![L]\!].$$

□

Definition 19 (Same-Tag Coverage Condition)**.** *For* $L, R \in \mathsf{STypes}$*, define:*

$$\mathsf{STC}(L, R) \iff \forall k \in \mathsf{tags}(L) \cap \mathsf{tags}(R).\ [\![L(k)]\!] \subseteq [\![R(k)]\!]$$

Lemma 20 (STypes Diff Satisfies (P2) under STC)**.** *If* $\mathsf{STC}(L, R)$ *holds, then* $[\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!] \cup [\![R]\!] \supseteq [\![L]\!]$*.*

Proof. Let $x \in [\![L]\!]$. Then $x \in [\![L(k)]\!]$ for some $k \in \mathsf{tags}(L)$.

- Case $k \notin \mathsf{tags}(R)$: Then $k \in \mathsf{tags}(\mathsf{diff}_{\mathsf{STypes}}(L, R))$ and $\mathsf{diff}_{\mathsf{STypes}}(L, R)(k) = L(k)$, so $x \in [\![\mathsf{diff}_{\mathsf{STypes}}(L, R)]\!]$.
- Case $k \in \mathsf{tags}(R)$: By STC, $[\![L(k)]\!] \subseteq [\![R(k)]\!] \subseteq [\![R]\!]$, so $x \in [\![R]\!]$.

□

Corollary 21 (STypes Diff under STC)**.** *If* $\mathsf{STC}(L, R)$ *holds, then (P2) holds. In particular, if* $L(k) = R(k)$ *for all* $k \in \mathsf{tags}(L) \cap \mathsf{tags}(R)$*, then (P2) holds.*

Lemma 22 (STypes Diff Completeness)**.** $\mathsf{diff}_{\mathsf{STypes}}$ *is complete (i.e.,* $[\![\mathsf{diff}_{\mathsf{STypes}}(L,R)]\!] = [\![L]\!] \setminus [\![R]\!]$*) if and only if* $\mathsf{STC}(L,R)$*.*

Proof. ($\Rightarrow$) Suppose completeness holds and let $k \in \mathsf{tags}(L) \cap \mathsf{tags}(R)$. Since $k \notin \mathsf{tags}(\mathsf{diff}_{\mathsf{STypes}}(L,R))$, the component $L(k)$ contributes no value to $[\![L]\!] \setminus [\![R]\!]$ under completeness, hence $[\![L(k)]\!] \subseteq [\![R]\!]$. By A1 and the "at most one entry per tag" invariant for STypes, $[\![L(k)]\!]$ can only overlap with $[\![R(k)]\!]$, so $[\![L(k)]\!] \subseteq [\![R(k)]\!]$.

($\Leftarrow$) We show $[\![\mathsf{diff}_{\mathsf{STypes}}(L,R)]\!] = [\![L]\!] \setminus [\![R]\!]$.

($\subseteq$) Let $x \in [\![\mathsf{diff}_{\mathsf{STypes}}(L,R)]\!]$. Then $x \in [\![L(k)]\!]$ for some $k \in \mathsf{tags}(L) \setminus \mathsf{tags}(R)$. By A1, $x \notin [\![R]\!]$ (no type in R has tag k), so $x \in [\![L]\!] \setminus [\![R]\!]$.

($\supseteq$) Let $x \in [\![L]\!] \setminus [\![R]\!]$. Then $x \in [\![L(k)]\!]$ for some $k \in \mathsf{tags}(L)$ and $x \notin [\![R]\!]$. If $k \in \mathsf{tags}(R)$, then by hypothesis $[\![L(k)]\!] \subseteq [\![R(k)]\!] \subseteq [\![R]\!]$, contradicting $x \notin [\![R]\!]$. So $k \notin \mathsf{tags}(R)$, meaning $k \in \mathsf{tags}(\mathsf{diff}_{\mathsf{STypes}}(L,R))$ and $x \in [\![\mathsf{diff}_{\mathsf{STypes}}(L,R)]\!]$. □

Corollary 23 (When STypes Diff is Incomplete)**.** $\mathsf{diff}_{\mathsf{STypes}}(L,R)$ *is incomplete when there exists* $k \in \mathsf{tags}(L) \cap \mathsf{tags}(R)$ *with* $[\![L(k)]\!] \not\subseteq [\![R(k)]\!]$*. Example:* $\mathsf{diff}_{\mathsf{STypes}}(\{4 \mapsto \mathit{number}\}, \{4 \mapsto \mathit{1}\}) = \emptyset$*, but* $[\![\mathit{number}]\!] \setminus [\![\mathit{1}]\!] \neq \emptyset$*.*

3.2 TInsts Diff

Definition 24 (TInsts Diff)**.**

$$\mathsf{diff}_{\mathsf{TInsts}}(L,R) = L \setminus R = \{i \in L \mid i \notin R\}$$

Note. *In the implementation,* TInsts *is represented as a collection of instance AST nodes and* $\mathsf{diff}_{\mathsf{TInsts}}$ *removes elements based on* structural equality *of instances. In this paper, that corresponds exactly to the usual mathematical set difference on the set* TInst *(i.e. membership/equality is the syntactic identity of instances, not semantic overlap).*

Lemma 25 (TInsts Diff Satisfies (P1) and (P2))**.** $[\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!] \subseteq [\![L]\!]$ *and* $[\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!] \cup [\![R]\!] \supseteq [\![L]\!]$*.*

Proof. (P1): $\mathsf{diff}_{\mathsf{TInsts}}(L,R) \subseteq L$, so $[\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!] \subseteq [\![L]\!]$.

(P2): Let $x \in [\![L]\!]$. Then $x \in [\![i]\!]$ for some $i \in L$.

- If $i \in R$: Then $x \in [\![i]\!] \subseteq [\![R]\!]$.
- If $i \notin R$: Then $i \in \mathsf{diff}_{\mathsf{TInsts}}(L,R)$, so $x \in [\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!]$.

□

Lemma 26 (TInsts Diff Completeness)**.** $\mathsf{diff}_{\mathsf{TInsts}}$ *is complete if and only if for all* $i \in L \setminus R$ *and* $j \in R$*:* $[\![i]\!] \cap [\![j]\!] = \emptyset$*.*

Proof. ($\Rightarrow$) Suppose completeness and let $i \in L \setminus R$ and $j \in R$. If $x \in [\![i]\!] \cap [\![j]\!]$, then $i \in \mathsf{diff}_{\mathsf{TInsts}}(L,R)$, so $x \in [\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!] = [\![L]\!] \setminus [\![R]\!]$. But $x \in [\![j]\!] \subseteq [\![R]\!]$, contradiction.

($\Leftarrow$) We show $[\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!] = [\![L]\!] \setminus [\![R]\!]$.

($\subseteq$) Let $x \in [\![\mathsf{diff}_{\mathsf{TInsts}}(L,R)]\!]$. Then $x \in [\![i]\!]$ for some $i \in L \setminus R$. For any $j \in R$, $[\![i]\!] \cap [\![j]\!] = \emptyset$ by hypothesis. Thus $x \notin [\![R]\!]$, so $x \in [\![L]\!] \setminus [\![R]\!]$.

($\supseteq$) Let $x \in [\![L]\!] \setminus [\![R]\!]$. Then $x \in [\![i]\!]$ for some $i \in L$. If $i \in R$, then $x \in [\![R]\!]$, contradiction. So $i \in L \setminus R = \mathsf{diff}_{\mathsf{TInsts}}(L,R)$. □

3.3 Clause Diff

Definition 27 (Clause Diff)**.** *For clauses* $c_1 = (\mathsf{vars}_1, \mathsf{stypes}_1)$ *and* $c_2 = (\mathsf{vars}_2, \mathsf{stypes}_2)$*:*

$$\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2) = \begin{cases} \bot & \text{if } \mathsf{stypes}_2 = \mathsf{None} \\ c_1 & \text{if } \mathsf{stypes}_1 = \mathsf{None} \\ (\mathsf{vars}_1, \mathsf{Some}(\mathsf{diff}_{\mathsf{STypes}}(S_1, S_2))) & \text{if } \mathsf{stypes}_1 = \mathsf{Some}(S_1), \mathsf{stypes}_2 = \mathsf{Some}(S_2) \end{cases}$$

where $\bot$ *represents the empty type (dropped from results) with* $[\![\bot]\!] = \emptyset$*.*

Lemma 28 (Clause Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2)]\!] \subseteq [\![c_1]\!]$*.*

Proof. Case analysis:

- $\mathsf{stypes}_2 = \mathsf{None}$: Result is $\bot$. $[\![\bot]\!] = \emptyset \subseteq [\![c_1]\!]$.
- $\mathsf{stypes}_1 = \mathsf{None}$: Result is c_1. $[\![c_1]\!] \subseteq [\![c_1]\!]$.
- Both Some: $[\![(\mathsf{vars}_1, \mathsf{Some}(\mathsf{diff}_{\mathsf{STypes}}(S_1, S_2)))]\!] = \left(\bigcap_{v \in \mathsf{vars}_1} [\![v]\!]\right) \cap [\![\mathsf{diff}_{\mathsf{STypes}}(S_1, S_2)]\!] \subseteq \left(\bigcap_{v \in \mathsf{vars}_1} [\![v]\!]\right) \cap [\![S_1]\!] = [\![c_1]\!]$.

□

Lemma 29 (Clause Diff Satisfies (P2) Conditionally)**.** $\mathsf{diff}_{\mathsf{Clause}}$ *satisfies (P2) when:*

- *Case* $\mathsf{stypes}_2 = \mathsf{None}$*: Always for well-formed clauses (since then* c_2 *is the unique top clause and* $[\![c_2]\!] = \mathsf{Val} \supseteq [\![c_1]\!]$*).*
- *Case* $\mathsf{stypes}_1 = \mathsf{None}$*: Always (since result is* c_1*).*
- *Case both* Some*: When* $\mathsf{diff}_{\mathsf{STypes}}(S_1, S_2)$ *satisfies (P2).*

3.4 Clauses Diff

Definition 30 (Clauses Diff)**.** *For* $C_1, C_2 \in \mathsf{Clauses}$*, let* $c_\emptyset = C_2(\emptyset)$ *if* $\emptyset \in \mathrm{dom}(C_2)$*, else undefined.*

$$\mathsf{diff}_{\mathsf{Clauses}}(C_1, C_2)(k) = \begin{cases} \mathsf{diff}_{\mathsf{Clause}}(\mathsf{diff}_{\mathsf{Clause}}(C_1(k), C_2(k)), c_\emptyset) & \text{if } k \in \mathrm{dom}(C_2) \wedge c_\emptyset \text{ defined} \\ \mathsf{diff}_{\mathsf{Clause}}(C_1(k), C_2(k)) & \text{if } k \in \mathrm{dom}(C_2) \wedge c_\emptyset \text{ undefined} \\ \mathsf{diff}_{\mathsf{Clause}}(C_1(k), c_\emptyset) & \text{if } k \notin \mathrm{dom}(C_2) \wedge c_\emptyset \text{ defined} \\ C_1(k) & \text{if } k \notin \mathrm{dom}(C_2) \wedge c_\emptyset \text{ undefined} \end{cases}$$

for $k \in \mathrm{dom}(C_1)$*. Clauses mapped to* $\bot$ *are dropped.*

Note. *Whenever* $k \in \mathrm{dom}(C_2)$*, the call* $\mathsf{diff}_{\mathsf{Clause}}(C_1(k), C_2(k))$ *compares clauses stored under the* same key k*, so their variable-sets match by the* Clauses *well-formedness condition (keys agree with clause variables). The only subtraction that may use a different key is via* $c_\emptyset = C_2(\emptyset)$*, which is deliberately applied to every* $C_1(k)$*; this corresponds to the implementation behavior and explains why the right-hand clause's variables are not used by* $\mathsf{diff}_{\mathsf{Clause}}$*.*

Why the $\emptyset$-clause matters. *The distinguished key $\emptyset$ corresponds to clauses that do not require any type parameters. When present on the right, subtracting $c_\emptyset$ from every left clause models a "global" elimination that should apply regardless of which type-parameter constraints produced the left clause.*

Lemma 31 (Clauses Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{Clauses}}(C_1, C_2)]\!] \subseteq [\![C_1]\!]$.

Proof. Each output clause is derived from $C_1(k)$ via $\mathsf{diff}_{\mathsf{Clause}}$, which satisfies (P1). □

3.5 TType Diff

Definition 32 (TType Diff)**.** *For $t_1 = (\mathsf{insts}_1, \mathsf{clauses}_1)$ and $t_2 = (\mathsf{insts}_2, \mathsf{clauses}_2)$:*

$$\mathsf{diff}_{\mathsf{TType}}(t_1, t_2) = (\mathsf{diff}_{\mathsf{TInsts}}(\mathsf{insts}_1, \mathsf{insts}_2), \mathsf{diff}_{\mathsf{Clauses}}(\mathsf{clauses}_1, \mathsf{clauses}_2))$$

Theorem 33 (TType Diff Satisfies (P1))**.** $[\![\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)]\!] \subseteq [\![t_1]\!]$.

Proof.

$$\begin{aligned}[\![\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)]\!] &= [\![\mathsf{diff}_{\mathsf{TInsts}}(\mathsf{insts}_1, \mathsf{insts}_2)]\!] \cup [\![\mathsf{diff}_{\mathsf{Clauses}}(\mathsf{clauses}_1, \mathsf{clauses}_2)]\!] \\ &\subseteq [\![\mathsf{insts}_1]\!] \cup [\![\mathsf{clauses}_1]\!] = [\![t_1]\!]\end{aligned}$$

by Lemmas 25 and 31. □

Theorem 34 (TType Diff Satisfies (P2) Conditionally)**.** $\mathsf{diff}_{\mathsf{TType}}$ *satisfies (P2) when all underlying* $\mathsf{diff}_{\mathsf{STypes}}$ *calls satisfy (P2). In particular, (P2) holds when* STC *holds for all same-tag eliminations.*

Theorem 35 (TType Diff Completeness)**.** $\mathsf{diff}_{\mathsf{TType}}$ *is complete when:*

1. $\mathsf{diff}_{\mathsf{TInsts}}$ *is complete (structurally distinct instances are semantically disjoint), and*
2. $\mathsf{diff}_{\mathsf{STypes}}$ *is complete for all clause pairs (same-tag implies $[\![s]\!] \subseteq [\![s']\!]$).*

Corollary 36 (Explicit Completeness Conditions for $\mathsf{diff}_{\mathsf{TType}}$)**.** *Let $t_1 = (\mathsf{insts}_1, \mathsf{clauses}_1)$ and $t_2 = (\mathsf{insts}_2, \mathsf{clauses}_2)$. Then $\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)$ is complete (i.e. $[\![\mathsf{diff}_{\mathsf{TType}}(t_1, t_2)]\!] = [\![t_1]\!] \setminus [\![t_2]\!]$) under the following explicit hypotheses:*

1. ***Instance disjointness across sides:*** *for all $i \in \mathsf{insts}_1$ and $j \in \mathsf{insts}_2$ with $i \neq j$, we have $[\![i]\!] \cap [\![j]\!] = \emptyset$.*
2. ***Top-clause alignment:*** *if $\mathsf{clauses}_1$ is top (i.e. it contains the unique top clause $(\emptyset, \mathsf{None})$), then $\mathsf{clauses}_2$ is also top.*
3. ***Same-tag coverage for every clause subtraction:***
   (a) *For every $k \in \mathrm{dom}(\mathsf{clauses}_1) \cap \mathrm{dom}(\mathsf{clauses}_2)$, if $\mathsf{clauses}_1(k) = (k, \mathsf{Some}(S_1))$ and $\mathsf{clauses}_2(k) = (k, \mathsf{Some}(S_2))$, then $\mathsf{STC}(S_1, S_2)$.*
   (b) *If $\emptyset \in \mathrm{dom}(\mathsf{clauses}_2)$ and $\mathsf{clauses}_2(\emptyset) = (\emptyset, \mathsf{Some}(S_0))$, then for every $k \in \mathrm{dom}(\mathsf{clauses}_1)$ with $\mathsf{clauses}_1(k) = (k, \mathsf{Some}(S_1))$, we have $\mathsf{STC}(S_1, S_0)$.*

4 Application to Type Inference

Definition 37 (Subtyping Check with Validation)**.** *When checking $u_1 \leq u_2$ where u_2 contains type variable T:*

1. *Compute* $\mathsf{diff}(u_1, u_2)$ *and grow T's lower bound.*
2. *Expand u_1, u_2 using the substitution.*
3. *Validate via* $\mathsf{isSub}(expand(u_1), expand(u_2))$*.*

Theorem 38 (Inference Soundness)**.** *If* isSub *correctly validates subtyping on expanded types, then type inference is sound regardless of whether* diff *satisfies (P2).*

Proof. The diff result is used only to guide inference (e.g. by accumulating bounds or hints about T). The acceptance/rejection decision is made solely by isSub on expanded types. Therefore, an imprecise diff can at worst cause incompleteness (e.g. by failing to discover a useful bound, leading to a later rejection), but it cannot cause an unsound acceptance when isSub is correct. □

5 Examples of Incompleteness

We present two concrete examples where diff is incomplete, i.e., $[\![\mathsf{diff}(t_1, t_2)]\!] \neq [\![t_1]\!] \setminus [\![t_2]\!]$ (it may be a strict subset or a strict superset of the true set difference).

5.1 Example 1: Same-Tag Literal Types

TypeScript code:

```
function f(x: 1 | 2): 2 {
  if (x === 1) { return 2; }
  return x;  // x should be narrowed to 2
}
```

Mathematical setup: Both `1` and `2` have the same runtime tag 4. Because STypes stores at most one SType per tag, it cannot precisely represent unions of same-tag literals such as `1 | 2`. In practice, the implementation merges same-tag literals into a single representative SType for the tag (often a wider type such as `number`), which loses the ability to subtract a single literal while keeping another.

Incompleteness: Since $\mathsf{tag}(\texttt{1}) = \mathsf{tag}(\texttt{2}) = 4$ and STypes stores at most one entry per tag, $\mathsf{diff}_{\mathsf{STypes}}$ cannot distinguish between literals of the same tag. For a simpler illustration:

$$\mathsf{diff}_{\mathsf{STypes}}(\{4 \mapsto \texttt{number}\}, \{4 \mapsto \texttt{1}\}) = \emptyset$$

But $[\![\texttt{number}]\!] \setminus [\![\texttt{1}]\!] =$ all numbers except $1 \neq \emptyset$.

The condition $\mathsf{STC}(L, R)$ fails: $[\![\texttt{number}]\!] \not\subseteq [\![\texttt{1}]\!]$.

5.2 Example 2: Top Type (unknown)

TypeScript code (requires `--strictNullChecks`):

```
function f(x: unknown): {} {
  if (x !== null && x !== undefined) {
    return x;  // x should be narrowed to {}
  }
  return {};
}
```

Mathematical setup: The type `unknown` is represented as a clause with $\mathsf{stypes} = \mathsf{None}$ (top). Let $c_1 = (\emptyset, \mathsf{None})$ represent `unknown`. Let c_2 represent `null | undefined`: $c_2 = (\emptyset, \mathsf{Some}(\{0 \mapsto \texttt{undefined}, 2 \mapsto \texttt{null}\}))$.

Incompleteness: By definition of $\mathsf{diff}_{\mathsf{Clause}}$:

$$\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2) = c_1 \quad (\text{since } \mathsf{stypes}_1 = \mathsf{None})$$

So $[\![\mathsf{diff}_{\mathsf{Clause}}(c_1, c_2)]\!] = [\![c_1]\!] = \mathsf{Val}$ (all values).

But the true set difference is:

$$[\![c_1]\!] \setminus [\![c_2]\!] = \mathsf{Val} \setminus ([\![\texttt{null}]\!] \cup [\![\texttt{undefined}]\!]) \neq \mathsf{Val}$$

The result is incomplete: diff returns `unknown` instead of the narrowed type $\{\}$ (non-nullish values). This is because the clause diff cannot "subtract from top"—it simply returns the left clause unchanged when $\mathsf{stypes}_1 = \mathsf{None}$.